# Research brief | AITR 7

Chinmay Thakur | MDE2026011 | IIIT Allahabad | Updated 21 October 2026

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| **Review and sources** | T23: citation integrity, with presentation, identity, access and claim support separated. Verified original sources: Abbonato (2026), Badalova & Mayr (2026), Shi et al. (2026), and Gao et al. (2023). |
| **Research gap** | The released checker decisions permit testing whether simple agreement reduces false alarms. This is a tractable inferred question, not a claim of a new voting algorithm or a systematic novelty result. |
| **Primary question** | Does a fixed three-of-five majority reduce RefChecker's false-positive rate by at least 0.10 absolute while retaining recall of at least 0.80 on the same corpus? |
| **Supporting questions** | How do thresholds, panel removals and source-held-out selection change the trade-off? What review workload and residual errors follow from fixed disagreement deferral? |
| **Pathway / build** | Controlled comparative reanalysis. Implement a strict data loader, pure voting/triage functions, grouped threshold selection, metrics, tests and reproducible evidence exports. Primary study uses recorded decisions. A 21 October supplement executes pinned CheckIfExist code on citation strings. |
| **Data and versions** | 104 references from three source documents, five checker flags each. Zenodo v1.0.0: 33 problematic / 71 verified. Author-corrected v1.1.0: 35 / 69. Both CC BY 4.0; raw bytes, metadata and hashes retained. |
| **Comparators / controls** | All five released checkers; k=1..5 thresholds; leave-one-document-out selection; five strict 3-of-4 removals. Same rows and labels within a release. Labels never enter fixed predictors; held-out labels never choose their own threshold. |
| **Measures / depth** | Primary recall, false-positive rate and balanced accuracy, with confusion counts. Threshold and panel sensitivity, per-document analysis, triage coverage, error patterns, and conditional paired bootstrap (5,000; seed 20261018). |
| **Feasibility / risks** | CPU-only tabular processing, no models or paid APIs. Small purposive corpus, inherited labels, heterogeneous sources, conservative duplicate screen, no human-trust or semantic-support measurements. |
| **Verified checkpoint** | Historical RefChecker matrix reproduced: TP32 / FP36 / TN35 / FN1. 16 core and 8 baseline tests pass; 24 primary numerical files reproduce. CrossRef component: 104/104 complete; full cascade: 23/104 complete. Corrected majority has 36 false alarms and 4 misses: the primary target fails. |
| **Provenance** | Protocol frozen before reference-level outcomes; published aggregates already known. A2 records the label correction; A3/A4 transparently add executable baseline checks after the primary study. Student review and live demonstration remain personal responsibilities. |

Official data: doi.org/10.5281/zenodo.21457492 and doi.org/10.5281/zenodo.23236264. Full source links, methods, evidence and amendments are included in the repository.